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Unlocking organizational creativity: Unique effects of corporate social responsibility in leveraging individual creativity☆

He Li[a,*], Philip J. Steinberg[b,*], Hille C. Bruns[c], Jordi A. Surroca[d]

[a] HeXie Management Research Centre and College of Industry-Entrepreneurs, Xi'an Jiaotong-Liverpool University, Chongwen Road 8, 215123 Suzhou, China
[b] Faculty of Economics and Business, University of Groningen, Nettelbosje 2, 9747 AE Groningen, the Netherlands
[c] Faculty of Social Sciences and Humanities, Vrije Universiteit Amsterdam, De Boelelaan 1105, 1081 HV Amsterdam, the Netherlands
[d] Management School, University of Liverpool, Chatham Street, Liverpool L69 7ZX, UK

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ABSTRACT

Individual creativity is considered a precursor to organizational creativity. However, organizational creativity is more than the sum of employee creative capabilities: It emerges through collaborative interaction among employees. We propose that corporate social responsibility (CSR) serves both as an internal driver and a context that supports the transformation of employee into organizational creativity. Specifically, we theorize that employee-related CSR enhances organizational creativity by boosting employee creativity and that non-employee-related CSR positively moderates the relationship between employee and organizational creativity. Building on a sample of 833 US firms from 2009 to 2019 and using a novel approach to measure organizational creativity, we find support for our theorization. Our theory and findings contribute to creativity literature by stressing the role of context in the transformation of employee into organizational creativity.

1. Introduction

Organizational creativity is considered an essential precursor to innovation (Anderson et al., 2014; Sarooghi et al., 2015). It refers to "the creation of a valuable, useful new product, service, idea, procedure, or process by individuals working together in a complex social system" (Woodman et al., 1993, p. 293). Employees, through their capacity to generate novel and useful ideas based on their individual creativity (Amabile et al., 1996; Amabile, 1988; Oldham and Cummings, 1996), are regarded as the foundation of organizational creativity (Amabile et al., 1996; Ghosh, 2015; Woodman et al., 1993). Hence, organizations must unleash their employees' creative potential to promote creative outcomes (Amabile, 1988; Woodman et al., 1993; Zhou and George, 2003). However, the upward transfer of employee to organizational creativity is increasingly difficult (Sarooghi et al., 2015). A burgeoning stream of research explores factors that may facilitate or inhibit that transformation (for a review, see Lua et al., 2024), such as individuals' leadership and entrepreneurial orientation (Gao et al., 2021), but this research does not fully account for organizational context.

This oversight is important because organizational creativity depends on individual context such as facilitation, rewards (Woodman et al., 1993), and work environment (Amabile and Pratt, 2016), as well as on organizational context such as human resource systems (Jeong and Shin, 2019) and organizational culture (Hempel and Sue-Chan, 2010). Clearly, organizational creativity is more than the sum of isolated creative abilities of employees (Hargadon and Bechky, 2006)—it materializes synergistically through collaborative interactions in which individuals exchange and develop creative ideas within an organizational context (e.g., Hargadon and Bechky, 2006; Woodman et al., 1993). Hence, beyond cognitive attributes, motivation and idea generation that nourish individual creativity (Amabile et al., 1996; Amabile, 1997), the context in which joint endeavors unfold is key to the transformation of individual to organizational creativity (Hargadon and Bechky, 2006; Jeong and Shin, 2019; Wu et al., 2022).

This study aims to enlarge our conceptual and practical understanding of how the context influences creativity and its transformation. With this aim in mind, we shift our attention to the role of corporate social responsibility (CSR). CSR encompasses a large array of activities—other than those required by law—with which organizations address social and environmental concerns vis-à-vis stakeholders (Aguinis and Glavas, 2012). For example, CSR fosters employee motivation and engagement (Chaudhary and Akhouri, 2018; Rupp et al.,

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* Corresponding authors.
E-mail addresses: He.Li@xjtlu.edu.cn (H. Li), p.j.steinberg@rug.nl (P.J. Steinberg), h.c.bruns2@vu.nl (H.C. Bruns), j.surroca@liverpool.ac.uk (J.A. Surroca).

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2006; Rupp et al., 2018; Simpson et al., 2020). CSR has been associated with high-commitment HR practices, a culture of innovation, mutual trust, and collaborative relationships among employees (Surroca et al., 2010); also, it makes individuals experience meaningfulness in their work (Aguinis and Glavas, 2019). Consequently, CSR constitutes a key contextual factor in organizations (Tuan, 2014).

Building on these ideas, we argue that CSR serves as a catalyst for the transformation of employee into organizational creativity. Among the different types of CSR (e.g., Hawn and Ioannou, 2016), employee-related CSR and CSR related to the environment and society at large are particularly relevant to our study. We theorize that CSR directed towards employees increases their creativity by inspiring them to assimilate new information and explore new ideas (Hur et al., 2018; Liu et al., 2017). We do not expect the same direct effect of non-employee-related CSR, however. Although CSR targeted towards other social and environmental goals does not directly benefit employees, it may still provide them with indirect, intangible benefits (Flammer and Kacperczyk, 2019). Non-employee-related CSR fosters a sense of affective commitment and identification with their organization (De Roeck and Delobbe, 2012; Farooq et al., 2017), which increases employees' engagement with their co-workers (Chaudhary and Akhouri, 2019; Rupp et al., 2018; Yang et al., 2020) and improves information sharing and collaboration (Dukerich et al., 2002). The latter, in turn, supports the transformation of employee into organizational creativity. Hence, we hypothesize that non-employee-related CSR moderates the relationship between employee creativity and organizational creativity.

To test our theorizing, we employed a novel approach to estimate organizational creativity by creating a novel and replicable text-based measure applied to the Management Discussion and Analysis (MD&A) sections of firms' 10-K reports. Based on a sample of 833 U.S. firms spanning 2009 to 2019, we show that employee-related CSR boosts organizational creativity by fostering employee creativity. Additionally, our findings reveal that non-employee-related CSR does not directly influence employee creativity but facilitates the transformation of employee into organizational creativity and amplifies the indirect influence of employee-related CSR on organizational creativity.

This research makes three contributions. First, this study contributes to the literature on the intricate relationship between individual and organizational creativity by exploring the role of CSR in this relationship. Our study addresses the question of which types of CSR facilitate the transformation of individual into organizational creativity and in what way. In doing so, we respond to calls to study this transformation (Gao et al., 2021; Lua et al., 2024). We extend prior research that has identified employee-level drivers of organizational creativity, such as outward personnel mobility networks (Godart et al., 2014) and high-performing work practices (Jeong and Shin, 2019) that organizations cannot control. Hence, our study puts forth CSR as a tool to help transform individual into organizational creativity. Second, this study contributes to the literature on CSR and creativity that has treated CSR as a monolithic construct (Abdelmotaleb et al., 2018; Brammer et al., 2015) by distinguishing employee-related from non-employee-related CSR. Hence, our study offers a more nuanced understanding of how specific types of CSR foster creativity at different levels. Lastly, this study contributes an innovative and replicable measure of organizational creativity based on a textual analysis of corporate annual reports. This is valuable because there is a dearth of measures of organizational creativity (Lua et al., 2024).

2. Theoretical background

2.1. Transformation of individual into organizational creativity and the role of context

There is an ongoing stream of literature on the transformation of creativity in organizations. Individual creativity provides raw material for novel and useful ideas, and interactions and exchanges among co-workers play a pivotal role in how this raw material evolves into organizational creativity (for a review, see Blomberg et al., 2017). The componential model of creativity and innovation in organizations (Amabile, 1988, 1997; Amabile and Pratt, 2016) specifies domain-relevant experience, creativity-relevant skills and processes, and intrinsic motivation as individual-level factors that further organizational creativity and innovation; this model also points to the role of context in terms of social and work environment that promotes creativity. Indeed, a variety of contextual factors fosters individual and organizational creativity. For example, the quality of exchanges between employees and supervisors (Liao et al., 2010), communication patterns, diversity, collaborative practices (Shalley and Gilson, 2004), organizational culture (Amabile, 1997), autonomy, encouragement from supervisors, and absence of excessive constraints (Amabile and Pratt, 2016) provide a supportive context for creativity. Our research examines which types of CSR, and through which mechanisms, provide a context that supports the transformation of individual into organizational creativity.

2.2. CSR informs context on the individual and organizational level

CSR is inextricably intertwined with the notion of stakeholders, i.e. those who influence or are influenced by the objectives of an organization such as employees, customers, community, and suppliers of capital and other resources (Freeman, 1984). CSR implies "*context-specific* organizational actions and policies that take into account stakeholders' expectations and the triple bottom line of economic, social, and environmental performance" (Aguinis, 2011, p. 855 *italics ours*). For example, the adoption of socially responsible practices informs organizational values, decision-making criteria, and procedures for working together (Aguilera et al., 2007; Howard-Grenville and Hoffman, 2003). Social and environmental considerations promote a culture of innovation and trust-based collaborative relationships with stakeholders (Sharma and Vredenburg, 1998). Hence, CSR shapes context via its ethical, social and environmental goals.

Several studies have highlighted the link between CSR and creative and innovative behaviors in organizations. Notably, CSR "is a construct that by its nature (Aguilera et al., 2007) connects the micro- (i.e., individual) and macro- (i.e., organizational) levels" (Hur et al., 2018, p. 631). At the individual level, CSR contributes to a strong moral identity of employees and a sense of psychological safety and meaningful work that fuels intrinsic motivation and commitment towards the organization, which empowers employees to experiment with new ideas and explore new approaches (Hur et al., 2018; Rupp et al., 2013). At the organizational level, CSR and creative outcomes are linked through the development of new business strategies (McWilliams and Siegel, 2000), organizational cultures (Du and Vieira, 2012), and ways of operating (Gao and Bansal, 2013) that enhance sustainability and financial performance. Below, we develop the argument that CSR encompasses a broad array of activities with diverse creative outcomes depending on the stakeholders towards which these activities are directed.

2.3. Employee-related CSR and non-employee-related CSR

Given that particular dimensions of CSR impact employees in the workplace, Flammer and Luo (2017) differentiated employee-related CSR from non-employee-related CSR depending on whether CSR activities provide direct benefits to employees or indirect benefits through activities that are targeted at other stakeholders but also matter to employees. Employee-related CSR provides direct benefits to employees (Flammer and Kacperczyk, 2019) in the form of fair wages, job security, diversity and safety in the workplace, continuous learning, training and development, health and work-life balance programs and participation in decision-making (Brammer et al., 2007; Farooq et al., 2017; Flammer and Kacperczyk, 2019; Rupp et al., 2006). Non-employee CSR activities target other stakeholders, such as local communities, suppliers, customers, the natural environment or the society at large. Non-employee-related CSR refers to environmental, social, and governance initiatives that do not directly impact employee well-being (Brammer et al., 2007; Chen et al., 2022; Cornelius et al., 2008). Examples include programs to engage with local communities and reduce firm's waste, energy consumption, and carbon footprint; the promotion of human rights and fair labor practices in the supply chain, and philanthropy (Flammer and Luo, 2017). Based on this typology of CSR activities and their differentiated effects on employees, in the next section we develop a framework that hypothesizes the relationships between (types of) CSR and creativity.

3. Hypotheses

3.1. The influence of CSR on employee creativity

According to the componential model of creativity and innovation, employees develop creative ideas in a stimulating environment. Hence, expertise in the task domain and skills in innovation management are not sufficient—employees need to feel motivated and free to explore creative solutions and assume risks (Amabile and Pratt, 2016). To facilitate this, organizations must find a proper balance between employees' intrinsic motivation and certain forms of extrinsic motivation, such as recognition, rewards, or constructive feedback (Amabile, 1993) and grant employees considerable autonomy (Liu et al., 2011) and psychological safety (Kark and Carmeli, 2009) for risk-taking. Providing employee motivational mechanisms as well as offsetting the personal risks associated with high unpredictability and long-term horizon of creative endeavors encourage employees' experimentation and trial-and-error (Flammer and Kacperczyk, 2016). Indeed, CSR initiatives increase employee motivation, engagement and creative performance (Bauman and Skitka, 2012; Hur et al., 2018). We argue that employee-related CSR that is directly relevant to employee satisfaction, engagement and motivation is more likely to boost individual creativity. Initiatives such as fair wages, job security and diversity and safety in the workplace create a positive work environment and enhance employee psychological and physiological well-being (Liu and Luo, 2022; Turker, 2009) which, in turn, enhances their job satisfaction, engagement with the organization, and intrinsic motivation to contribute to its success (Chaudhary and Akhouri, 2018; Rupp et al., 2006; Rupp et al., 2018; Simpson et al., 2020).

According to social exchange theory (Blau, 1964), there is a norm of reciprocity, so employees tend to reciprocate offerings made by their employers. Specifically, employee-related CSR fosters a trust-based relationship between employees and their organization. Creative ideas and innovative behaviors often depart from established routines and the status quo, which is inherently linked to taking risks (Rupp et al., 2018). Employees in organizations with a trust-based relationship are more inclined to take risks, as they believe their organizations will support their risky attempts regardless of the outcomes (Schoorman et al., 2007; Scott and Bruce, 1994). As firms with high employee-related CSR are more likely to foster trust among their employees, this will encourage employees to explore creative ideas and undertake innovative endeavors, ultimately leading to higher levels of employee creativity. In contrast, we don't expect the same degree of influence of non-employee-related CSR on individual creativity because that type of CSR is targeted at other stakeholders and does not evoke reciprocity. Thus, we hypothesize the following:

Hypothesis 1. Employee-related CSR is positively related to employee creativity.

3.1.1. Employee-related CSR, employee creativity, and organizational creativity

In the previous section, we have argued that employee-related CSR boosts employee creativity (Hypothesis 1), which in turn is widely recognized in the literature as a precursor to organizational creativity, as explained in 2.1. Based on these expectations, it can be argued that employee-related CSR is connected to organizational creativity through the mediation of employee creativity. Therefore, we anticipate that employee-related CSR—aimed at bolstering employee trust, cultivating positive psychological experiences, and fostering favorable attitudes towards the organization—will boost employee creativity. Earlier, we established that employee creativity is positively linked to organizational creativity. Thus, we propose the following mediation hypothesis:

Hypothesis 2. The relationship between employee-related CSR and organizational creativity is positively mediated by employee creativity.

3.2. Influence of CSR on organizational creativity

Although we are not aware of research that has examined the impact of CSR on organizational creativity (see Winchester and Medeiros, 2023), studies in related areas point to two distinct mechanisms through which CSR initiatives may influence organizational creativity. The first mechanism relates to the development of employee creativity. Accordingly, CSR can enhance organizational creativity by creating a fair, rewarding and supportive work environment that enhances employees' commitment, satisfaction, and sense of physical and psychological safety, which boosts their motivation to engage in creative endeavors. This will result in increased creativity outcomes at the organizational level (Hou et al., 2019; Hur et al., 2016). This mechanism is reflected in our Hypothesis 2.

The second mechanism relates to the conditions that tap into the potential of creative employees to work together to generate organizational outcomes. For example, when employees develop a sense of shared purpose within the organization, find meaningfulness and experience compassion at work, and trust their employer and colleagues, they become more likely to develop deeper affective commitment and identification with their organization and engage in corporate citizenship behaviors that support the organization's objectives, providing indirect benefits for the other employees (Amabile and Pratt, 2016; Blomberg et al., 2017; West and Sacramento, 2023). Though studies at the organizational level have not directly mentioned CSR as the driving force for these positive emotions, substantial research proposed that employees satisfy psychological needs of meaningfulness, sense of purpose, and trust from their firm's engagement in CSR (Aguilera et al., 2007; Aguinis and Glavas, 2012; Pratt et al., 2013; Rupp et al., 2013).

As explained previously, employee-related and non-employee-related CSR influence employee behavior through two different mechanisms. While employee-related CSR is positively related to employee satisfaction, engagement and motivation, which directly impact employees' individual creativity (Hypothesis 1), non-employee-related CSR instills in employees a sense of belonging and meaningfulness, even when they are not subject of these actions (Rupp et al., 2006). We claim that non-employee CSR activities are more likely to contribute to the second mechanism through which CSR positively influences organizational creativity. While employees do not receive direct benefits from non-employee-related CSR (Flammer and Luo, 2017), they indirectly benefit by deriving meaning from the fact that their employer takes such initiatives (Rupp et al., 2011). For example, indirect benefits result in elevated levels of employee engagement with their co-workers (Chaudhary and Akhouri, 2019; Rupp et al., 2018; Yang et al., 2020). Employees can find meaningfulness through serving the greater good (Aguinis and Glavas, 2019; Pratt et al., 2013; Rupp et al., 2013), which CSR symbolizes. Individuals constantly seek meaningfulness in their lives and frequently find it through their work (Aguinis and Glavas, 2019; Rosso et al., 2010). Meaningfulness refers to the degree of significance something holds for an individual (Pratt and Ashforth, 2003). Specifically, employees' perception of social impact and social worth profoundly influences their sense of significance and purpose (Grant, 2008).

Pratt and Ashforth (2003) put forward that employees find meaningfulness not just "in work" but also "at work". Tasks that contribute to social responsibility, such as volunteer projects, likely offer employees a sense of meaningfulness "in work" (Aguinis and Glavas, 2013; Carnahan et al., 2017; Pratt and Ashforth, 2003). Even when individual tasks do not directly contribute to societal or environmental betterment, employees can still find meaning "at work" by working for a socially responsible organization (Aguinis and Glavas, 2013; Carnahan et al., 2017; Pratt and Ashforth, 2003).

In socially responsible organizations, employees take greater pride in their association with their workplace, fostering a sense of meaningfulness in their work and at work (Chalofsky and Krishna, 2009; May et al., 2004; Ruck et al., 2017). By experiencing more meaningfulness, employees develop a sense of belonging to the group and greater identification with the organization's goals and practices. As a result, employees may show positive organizational citizenship behaviors and stronger commitment towards their coworkers (Farooq et al., 2017; Flammer and Luo, 2017; Rupp et al., 2006; Rupp et al., 2013; Rupp et al., 2011). Such employees are more likely to contribute to the effectiveness of their organizations in interactions with their colleagues (Pratt and Ashforth, 2003; Rupp et al., 2018) through sharing, building upon, and critiquing ideas (Aguinis and Glavas, 2019; Chaudhary and Akhouri, 2018). These interactions generate creative and valuable insights for organizations. Consequently, elevated levels of employee interaction with their co-workers facilitate the upward transfer of employee to organizational creativity.

In contrast, employees in organizations that rarely engage in non-employee-related CSR might lack motivation to engage in organizational citizenship and interact with coworkers and team members, limiting organizational creativity. The key role of collaborative interactions among individuals has been largely acknowledged in research on the transformation of individual to organizational creativity. Woodman et al. (1993), for example, explain that group creativity relies on—but is not merely the sum of—the individual creativity of group members; so that, while some novel insights within organizations may indeed originate from a single individual, most organizational creative outcomes emerge from collaborative interactions among employees—which non-employee CSR promotes (e.g., Chaudhary and Akhouri, 2019). Consequently, we anticipate that non-employee-related CSR strengthens the relationship between employee and organizational creativity, as hypothesized next:

Hypothesis 3. Non-employee-related CSR positively moderates the relationship between employee creativity and organizational creativity.

3.2.1. *The moderated mediation between CSR and creativity*

Integrating the preceding arguments leads us to propose a moderated mediation framework of employee-related CSR, employee creativity, non-employee-related CSR, and organizational creativity. Prior research suggests two mechanisms through which CSR supports the transformation of individual into organizational creativity. The first mechanism (Hypothesis 2) comprises employee-related CSR fostering a fair, rewarding and supportive work environment that enhances employee satisfaction, engagement and motivation, which boosts their willingness to engage in creative endeavors. The second mechanism comprises non-employee-related CSR that provides employees with a sense of shared purpose and meaningfulness at work. We expect the benefits of employee-related CSR, which enhance organizational creativity through employee creativity, to be especially pronounced when non-employee-related CSR is high, as it positively moderates the relationship between employee and organizational creativity. To state it formally:

Hypothesis 4. The strength of the mediated relationship between employee-related CSR and organizational creativity (via employee creativity) depends on non-employee-related CSR; the indirect and total effects of employee-related CSR on organizational creativity will be stronger with increasing non-employee-related CSR.

Fig. 1 depicts the hypothesized theoretical framework in this research.

4. Methods

4.1. *Sample selection and data*

The sample for this study consists of high-tech companies in the U.S. We focus on high-tech companies due to the heightened importance of creativity in high-tech industries compared to others. Moreover, the U.S. has the highest prevalence of high-tech companies and provides an excellent setting for empirical studies in terms of data availability. We identify high-tech industries using the Thomson Reuters classification of high-tech industries and the Thomson Reuters Business Classification (TRBC) code. Following previous research (Cloodt et al., 2006; Isaksson et al., 2016), the list of industries in the sample consists of chemicals (5110), industrial goods (5210), media & publishing (533020), healthcare services & equipment (5610), pharmaceuticals & medical research (5620), technology equipment (5710), software & IT services (5720), and telecommunications services (5740). We restricted our observation period to 2009–2019 due to the fragmented nature of the CSR data before 2009.

We collected our data from four databases. The CSR data were collected from Thomson Reuters' ASSET4 on Eikon. Thomson Reuters' ASSET4 is considered one of the most reliable sources of CSR information (Stellner et al., 2015). ASSET4 provides extensive and systematic CSR data on over 5000 global companies from publicly available sources, such as sustainability CSR reports and firm websites. Second, we drew data from the COMPUSTAT database, which offers detailed financial information on all publicly traded U.S. companies. We further

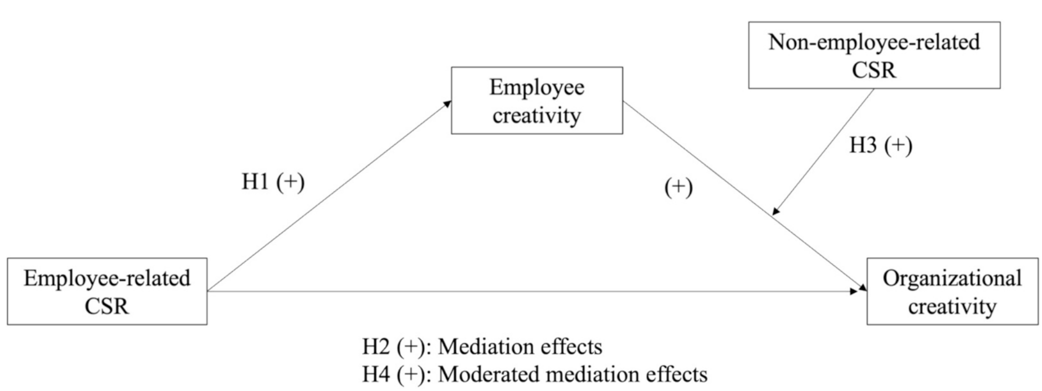

Fig. 1. Conceptual framework.

draw on 10-K filings from the U.S. Securities and Exchange Commission (SEC) EDGAR online database and the Notre Dame Software Repository for Accounting and Finance (SRAF) website to measure organizational creativity. Lastly, we used Nexis Uni, a database covering news, law, and business, to collect press releases and news related to employee creativity in firms.

To construct our sample, we selected all high-tech firms listed in ASSET4 from 2009 to 2019, yielding 5359 observations from 907 firms. Subsequently, these firms were matched with COMPUSTAT, reducing our database to 906 firms with 5332 observations. Following the exclusion of observations with missing values for our model variables, our final sample comprised unbalanced panel data, including 833 firms with 4667 firm-year observations spanning 2009 to 2019. We conducted a mean-comparison test (*t*-test) to compare our final sample with the 5359 observations from the ASSET4 database, checking for potential sample selection bias. The results, shown in Table A1 of the Online Appendix, indicated no statistically significant differences for all ASSET4 variables between the two samples, suggesting that sample selection bias does not seem to be of great concern.

4.2. Measures

4.2.1. Organizational creativity

Creativity researchers have employed different methods to measure organizational creativity. A typical method is the survey (e.g., Giustiniano et al., 2016; Lang and Lee, 2010), which allows researchers to measure creativity directly—though sacrificing generalizability and applicability outside its specific context. Another method to assess creativity is by means of patents. This method offers accessible, comparable, and objective information to measure creativity, but it may overlook significant creative endeavors as organizational creativity entails the generation of novel and practical ideas which is not necessarily fully captured in patents (Anderson et al., 2014; Bellstam et al., 2021). Moreover, while a few industries like technology and pharmaceuticals have high patenting rates, in most industries the majority of innovations are not patented, with only four sectors patenting more than half of their innovations (Arundel and Kabla, 1998). Thus, a measure relying on patents is not necessarily generalizable to a wide range of industries. Consequently, focusing exclusively on patents may lead to the exclusion of certain creative activities within firms. To address these concerns, we sought to measure organizational creativity through a computer-aided text analysis (CATA) of the information contained in the Management Discussion and Analysis (MD&A) section of firms' 10-K reports. The MD&A section reflects managerial perspectives and intentions (Caserio et al., 2019) and is therefore a useful source of information to capture organizational creativity. Moreover, all publicly listed firms in all sectors are legally required to publish 10-K reports, allowing for a wide application of our measure.

We developed a dictionary of organizational creativity in several steps. First, we identified potential keywords for our dictionary from academic articles. Specifically, the first author drew from the definition of creativity as "the creation of a valuable, useful new product, service, idea, procedure, or process by individuals working together in a complex social system" (Woodman et al., 1993, p. 293) as well as on previously validated scale items from Giustiniano et al. (2016), Jia et al. (2014), Lang and Lee (2010), and Moultrie and Young (2009). The other authors reviewed the word list and collaboratively determined which keywords to include. Appendix 1 (first column) shows the original list of extracted keywords from the definition and scale items, such as "novel" and "original".

Second, in line with McKenny et al. (2018), we refined our dictionary in two steps, using an inductive followed by a deductive approach. Initially, we employed an inductive approach, involving two independent human coders, both academic experts on creativity. Each coder reviewed 100 randomly selected MD&A sections from our sample (200 in total) to identify further dictionary words that may approximate firms' organizational creativity based on our previously mentioned definition. Afterward, they discussed all potential dictionary words and collectively decided on the new words to be added to the list. This led to a list of 19 additional keywords, displayed in Appendix 1 (column 2).

Third, we followed a deductive approach and assessed every dictionary word resulting from step 1 and 2 in its context to evaluate its usefulness for measuring organizational creativity in the MD&A texts. We utilized WordStat to examine one hundred instances for each dictionary word, drawing on a new set of randomly selected MD&A sections (McKenny et al., 2018). One of the authors reviewed all instances and identified words with extremely low frequency or those that did not accurately capture organizational creativity as they were mostly used in different contexts, such as "change" and "restructuring". We then collectively discussed these words and decided whether to retain or remove them. This process resulted in the removal of 24 words largely unrelated to organizational creativity in the texts, leaving a final list of 25 word stems specifically associated with it. The final dictionary is presented in Appendix 1 (column 3). To illustrate how dictionary words relate to organizational creativity, we include examples demonstrating the use of various words in the MD&A texts in Appendix 2.

Finally, the creation of the word list enabled the use of WordStat software to count the dictionary words within all MD&A texts in our full sample. The score of *organizational creativity* was calculated by dividing the count of organizational creativity keywords by the count of total words.

In order to validate our newly developed measure of organizational creativity, we conducted a comprehensive analysis of its correlations with key indicators of innovation: R&D intensity, patent ratios, granted patents, economic importance of an innovation, technological impact and novelty, and Fortune's ranking on innovativeness. First, we investigated its relationship with R&D intensity, which is calculated by dividing the total expenditure on R&D by the firm's total assets (Hitt et al., 1997). We found a positive correlation between our dictionary-based organizational creativity score and R&D intensity ($r = 0.47$; $p < 0.01$). Second, we examined the correlations between our organizational creativity score and patent ratios. Patent data were obtained from the U.S. patent database via WRDS, covering the years 2011 to 2019. Recognizing that organizational creativity takes time to produce measurable outcomes, we anticipated a delayed effect. Consequently, we focused on the patent-to-total assets ratio and the patent-to-employee number ratio in year $t + 1$ (Heeley et al., 2007). The correlations were $r = 0.13$ ($p < 0.01$) and $r = 0.16$ ($p < 0.01$), respectively. Third, we analyzed the application year of granted patents from the WRDS US Patents dataset from 2009 to 2019. We aggregated the patent information at the firm-level by counting the number of granted patent applications in a given year per firm. Our analysis revealed a correlation between our organizational creativity score and granted patent applications in year t, with $r = 0.05$ ($p < 0.01$). Fourth, we assessed whether our score is related to Kogan et al.'s (2017) measure of economic importance of an innovation, using their data. Assessing that measure one year after the filing year and aggregating patent data at the firm-level (using the mean value of the patent-level measure), we find a correlation between our score and the economic importance of an innovation both in real value ($r = 0.10$, $p < 0.01$) and nominal value ($r = 0.09$, $p < 0.01$). Fifth, we also examined the correlation between our score and the patent-based measures of technological impact and novelty suggested by Arts et al. (2021). After aggregating these measures at the firm level and assessing them at $t + 1$, we found correlations of $r = 0.13$ ($p < 0.01$) for technological impact and $r = 0.31$ ($p < 0.01$) for technological novelty. Finally, we analyzed the relationship between our measure and Fortune's ranking of America's Most Innovative Companies in 2023. Given that the ranking list was available only for 2023, we analyzed the correlation between our 2019 organizational creativity score (the latest year in our sample) and the reverse of the ranking in 2023 (i.e., so that a higher score indicates higher organizational creativity). We found a strong correlation of $r = 0.56$ ($p < 0.05$) based on a sample size of 56 firms. These findings (for an overview see Appendix 3) suggest solid validity and accuracy of our organizational creativity measure.

4.2.2. Employee creativity

To approximate employee creativity, we collected information on awards for outstanding creative work received by a firm's employees in a given year. The use of awards as indicators of creativity is an established practice in creativity research (Caird, 1994; Cattani and Ferriani, 2008; Feist and Barron, 2003; Simonton, 2004). For example, Cattani and Ferriani (2008) employ awards and nominations as indicators to gauge individual creativity within the Hollywood film industry. Similar to organizational creativity, employee creativity is often estimated through surveys (e.g., Černe et al., 2014; Gong et al., 2013) or experiments (e.g., Kim and Kim, 2020). However, both methods lack generalizability and applicability beyond their specific contexts due to the controlled and artificial settings in which they are conducted. Additionally, they may suffer from participation bias, as participants are often less wealthy, have more leisure time, are interested in experimental tasks, and exhibit more prosocial behaviors such as volunteering (Slonim et al., 2013). This can lead to non-representative participant samples. Moreover, conducting surveys to assess employee creativity across a large number of public firms, particularly larger ones, would be impractical in our context due to the scale and limited responsiveness of our sample firms. Lastly, our measure captures a broader range of highly creative employees than one would measure based on patenting activity, as awards are also granted in industries with traditionally low patent activity and to professionals who may not typically be listed as inventors on patents.

To identify the awards, we initially conducted an article search using the Nexis-Uni database, one of the most widely used news archives in the social sciences. We refined our search to professional magazines and journals because they report industry-recognized awards and to narrow down results to a quantity that could be manually assessed in a next step. To keep our search strategy as broad as possible and to capture as many awards as possible, we used the keywords "company name" and "award" in our search. The word "prize" was not included because we found it consistently appears alongside "award", as in "award prize." Our search yielded 94 awards. To include as many awards as possible and to ensure that we did not miss any relevant awards, we subsequently checked the awarding institutes of the awards we had identified via our Nexis-Uni search for additional employee-related awards on the official website of each institute. This process yielded 1352 awards.

Next, we further filtered these awards. Specifically, we adopted several criteria to identify awards for inclusion in our study. First, the included awards must be offered by professional societies; company-created awards are excluded to prevent bias. Such awards could skew the results towards firms that implement more internal recognition programs, particularly larger companies or those with strong employee-related CSR, which may also create a correlation with our employee-related CSR measure, as these companies are more likely to reward employees internally. In general, professional societies reflect the more objective judgments of many experts who recognize and award individuals and organizations according to their achievements and contributions. Second, since our study focuses on employee creativity, awards for CEOs, founders, or other senior executives were excluded. The third criterion required the awards to be offered to corporate employees. Therefore, the awards given to academic staff not employed in our sample firms were excluded.

This search and selection process resulted in 160 awards. In a next step, we manually coded the award description to determine whether the awards relate to creative performance of the employees. Two independent coders scored each award on a scale of 0 to 2 based on the definition of creativity, "the generation of new and potentially valuable ideas concerning new products, services, manufacturing methods, and administrative processes" (Zhou and George, 2001, p. 682). A score of 0 indicates no relevance to creative performance, 1 signifies indirect relevance to creative performance, and 2 suggests strong relevance to creative performance. The inter-rater reliability is sufficiently high, as indicated by a correlation between the two coders' results of 0.87. Coders discussed all the disputed awards and reached an agreement on the scoring. Awards with a score of 0 were excluded from the final selection. Following the exclusion of awards without relevance to creative performance, we obtained a total of 104 awards. Examples of individual creativity awards are presented in Appendix 4. Finally, *employee creativity* is coded as 1 if at least one company employee received an award in a given year, and 0 otherwise.

We validated employee creativity using disambiguated inventor data from 2009 to 2019, sourced from the PatentsView database. Since the data lacks assignee information, we linked it to our dataset using the U.S. Patents (WRDS) and DISCERN databases (Arora et al., 2024a; Arora et al., 2024b). Inventors serve as proxies for creative employees, assuming more inventors indicate higher employee creativity. By matching disambiguated inventors to institutions using patent identifiers, we counted the number of inventors per firm. The correlation between the number of inventors in a firm and our employee creativity measure is $r = 0.18$ ($p < 0.01$).[1] Additionally, using inventor data from 2000 to 2019 from the same database, we examined the number of star inventors within firms (Moretti and Wilson, 2017). This method of identifying highly creative individuals within the firm aligns well with our approach, which focuses on outstanding employees who have received awards. We used two alternative operationalizations regarding star inventors. First, we defined star inventors as those within the top 10 % of all inventors by patent count over a 10-year rolling window. We calculated the number of star inventors in each firm, finding a correlation of $r = 0.15$ ($p < 0.01$) with employee creativity. Second, we defined star inventors as those in the top 10 % of all inventors based on patent citation counts from 2011 to 2019, obtained from the U.S. Patents (WRDS) database, including citations within the first two years of the patent grant. The two measures correlate at $r = 0.13$ ($p < 0.01$) between the number of star inventors in a firm and employee creativity. Again, the findings (for an overview see Appendix 3) provide evidence for the validity of our measure.

4.2.3. Employee-related CSR and non-employee-related CSR

The CSR data are obtained from the Thomson Reuters ASSET4 database, which is widely used in CSR studies (Cai et al., 2012; Cheng et al., 2014; García-Martínez et al., 2019). The ASSET4 database has a pyramid structure: 400 sub-indicators are grouped into ten categories, which, in turn, inform three pillars: environmental (emissions, resource use, environmental innovation), social (workforce, human rights, community, and product responsibility), and governance (management, shareholders, and CSR strategy). Each of these 10 categories is assessed on a scale from 0 to 100. The workforce category contains the four dimensions of employment quality, health and safety, training and development, and diversity. Following Flammer and Kacperczyk (2019), we measure *employee-related CSR* with the workforce category score. Moreover, *non-employee-related CSR* is measured by computing the average scores from all other categories within the governance, environmental, and social pillars, while excluding the workforce category.

4.2.4. Control variables

In the regression analyses testing the effect of employee-related CSR on employee creativity, we include control variables that according to prior research likely affect the latter. To control for labor proficiency, we include *employee productivity*, which is measured by sales per employee. We also control for firm-level factors, such as *firm size* and *firm performance*. We measure firm size as the number of full-time employees reported in COMPUSTAT. We operationalize firm performance as Tobin's

[1] Since both patents and awards are subject to a time lag in relation to employees' creative performance, we do not account for an additional time lag.

Q, which is the ratio of the market value of total assets to the book value of total assets. Lastly, we control for *non-employee-related CSR* because CSR may generally be a predictor of employee creativity.

In the regression analysis for *organizational creativity*, we control for additional factors that are likely to influence organizational creativity. *Slack resources* that Greve (2003) and others have shown to be influential for organizational creativity were operationalized as the current ratio (current assets/current liabilities). We also control for *human resource (HR) slack*, as human resources are likely to affect the relationship between employee creativity and organizational creativity. We measure HR slack as the ratio of employee number to plant size, subtracting the same ratio for industry (Lecuona and Reitzig, 2014). To partial out the possible effects of corporate strategy on organizational creativity, we controlled for the *exploration orientation* and *differentiation strategy*. We employed the CATA approach suggested by Uotila et al. (2009) to measure exploration orientation using the authors' word list with firms' 10-k documents. The differentiation strategy measures the ratio of general, selling, and administrative expenses to total sales (Berman et al., 1999). This measure reflects a firm's efforts to differentiate itself from others. Finally, we used year and industry dummies (two-digit TRBC code) to control for the effects of time and sector.

4.3. Analysis

To test our hypotheses, we used random-effects panel models to account for the panel characteristics of our data. We defined our panel using firm-year observations. To test Hypothesis 1 and 3, we used random-effects panel models. To test Hypothesis 1, we employed random-effects logistic regression models to account for the binary nature of the dependent variable, employee creativity. To test Hypothesis 3, we employed linear random-effects GLS models. Random-effects models are more appropriate than their fixed-effects counterparts in our setting for two main reasons. First, the adoption of a fixed-effects model is not appropriate given the relatively short and unbalanced panel structure of our data (Steinberg et al., 2017), as each firm is observed only 5.6 times on average. Secondly, the main variables in our model are relatively stable and time-invariant during the observed time period, especially employee-related CSR and organizational creativity. Fixed-effects models may yield biased estimates in such settings (Jensen and Zajac, 2004), which makes a random-effects model that accounts for between-firm effects more appropriate (Zhang et al., 2008). To test Hypothesis 2, we employed the method outlined by Baron and Kenny (1986) and utilized the Sobel test that examines the indirect effect of an independent variable on the dependent variable through the mediator (MacKinnon et al., 2002; Preacher and Hayes, 2004; Sobel, 1982). We also followed current mediation studies (e.g., Calic and Mosakowski, 2016) and employed the general decomposition method developed by Karlson and Holm (2011)–the Karlson-Holm-Breen (KHB) method–to determine the extent to which employee creativity mediates or explains the relationship between employee-related CSR and organizational creativity. Finally, to test Hypothesis 4, we employed the moderated path analytic approach (Edwards and Lambert, 2007; Jia et al., 2014; Liao et al., 2010; Liu et al., 2011) that integrates moderated regression procedures into the path analytic method for testing mediation. All variables were standardized before the model analysis, except for the dichotomous variables, year, and industry.

5. Results

Table 1 provides descriptive statistics and pairwise correlations for all variables. It is worth noting that there is a relatively high correlation of 0.64 between employee-related CSR and non-employee-related CSR. However, it is common for different types of CSR to exhibit high correlation (Hawn and Ioannou, 2016). The mean and individual variance inflation factors are smaller than 5 (Belsey et al., 1980), indicating no apparent problems of multicollinearity between the variables.

The results of our hypotheses tests appear in Table 2. Models 1 and 2 predict employee creativity, while Models 3 to 6 predict organizational creativity. Hypothesis 1 predicted a positive relationship between employee-related CSR and employee creativity. Model 1 predicts employee creativity using only the control variables. Model 2 adds employee-related CSR as a predictor. Results provided in Model 2 indicate that employee-related CSR ($\beta = 0.906$, $p < 0.01$) is positively and significantly related to employee creativity. Therefore, Hypothesis 1 is supported.

Next, we followed Baron and Kenny's (1986) procedure to test the hypothesized mediation effect (Hypothesis 2). Baron and Kenny (1986) outlined three prerequisites for a mediating effect. First, there should be a significant relationship between the independent variable (employee-related CSR) and the dependent variable (organizational creativity). Second, the independent variable (employee-related CSR) must be significantly associated with the mediator (employee creativity). Third, when the independent variable predicts the dependent variable, the mediator should display a significant relationship with the dependent variable. When these three prerequisites are satisfied, the effects of the independent variable on the dependent variable must be diminished in the third condition compared to the first. Hypothesis 2 posits that employee creativity positively mediates the relationship between employee-related CSR and organizational creativity. The combined results from Models 2, 4, and 5 in Table 2 do not seem to support a full mediation. Instead, our findings suggest two mechanisms from employee-related CSR to organizational creativity–a direct one and an indirect one through employee creativity.

To further examine our Hypothesis 2, we conducted the Sobel test, utilizing the bootstrapping approach (Hayes, 2013; Preacher and Hayes, 2004; Sobel, 1982). The results of the Sobel test revealed that the

Table 1
Descriptive statistics and correlations.

Variables	Mean	SD	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) Organizational creativity (ln)	1.85	0.50	1.00									
(2) Employee creativity	0.02	0.15	0.12***	1.00								
(3) Employee-related CSR	42.42	26.18	0.12***	0.21***	1.00							
(4) Non-employee-related CSR	35.30	18.98	−0.05***	0.20***	0.64***	1.00						
(5) Firm size (ln)	7.75	1.69	−0.22***	0.23***	0.51***	0.63***	1.00					
(6) Tobin's Q	2.84	2.20	0.24***	−0.01	−0.00	−0.18***	−0.33***	1.00				
(7) Employee productivity	0.01	1.12	−0.07***	0.01	−0.01	0.04*	0.08***	−0.02	1.00			
(8) Exploration orientation	0.53	0.21	0.21***	0.05***	0.11***	−0.04***	−0.02	0.13**	0.02	1.00		
(9) Differentiation strategy (ln)	−1.41	1.11	0.42***	0.00	−0.01	−0.27***	−0.40***	0.29***	−0.16***	0.19***	1.00	
(10) Slack resources (ln)	1.27	0.50	0.38***	−0.04***	−0.16***	−0.25***	−0.41***	0.19***	0.04***	0.11***	0.30***	1.00
(11) Human resource slack	0.00	0.00	0.05***	0.01	0.02	−0.03**	−0.05***	0.01	−0.03**	0.01	0.25***	0.07***

Notes: N = 4667; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 2
Results of hypotheses tests.

	Employee creativity Random-effects logistic regression		Organizational creativity Random-effects linear regression			
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Employee-related CSR		0.9055***		0.0170**	0.0169**	0.0174**
		(0.2720)		(0.0075)	(0.0075)	(0.0075)
Employee creativity					0.0464*	−0.0151
					(0.0255)	(0.0379)
Employee creativity X Non-employee-related CSR						0.0516**
						(0.0234)
Control variables						
Non-employee-related CSR	0.3224	−0.1078	0.0146*	0.0063	0.0064	0.0059
	(0.2483)	(0.2696)	(0.0082)	(0.0091)	(0.0091)	(0.0091)
Firm size	2.3094***	1.9939***	−0.0699***	−0.0764***	−0.0778***	−0.0778***
	(0.3769)	(0.3687)	(0.0141)	(0.0143)	(0.0143)	(0.0143)
Tobin's Q	2.6477**	1.9328	−0.0032	−0.0075	−0.0065	−0.0062
	(1.1686)	(1.2705)	(0.0335)	(0.0336)	(0.0336)	(0.0336)
Employee productivity	−0.7525	−0.5987	−0.0071*	−0.0071*	−0.0072**	−0.0071*
	(0.6125)	(0.6082)	(0.0036)	(0.0036)	(0.0036)	(0.0036)
Exploration orientation			0.0013	0.0012	0.0011	0.0012
			(0.0057)	(0.0056)	(0.0056)	(0.0057)
Differentiation strategy			0.0526***	0.0518***	0.0521***	0.0528***
			(0.0082)	(0.0082)	(0.0082)	(0.0083)
Slack resources			0.0677***	0.0678***	0.0685***	0.0679***
			(0.0074)	(0.0075)	(0.0075)	(0.0075)
Human resource slack			−0.0070	−0.0069	−0.0070	−0.0071
			(0.0044)	(0.0044)	(0.0044)	(0.0044)
Constant	−8.4006***	−7.6962***	1.6490***	1.6632***	1.6639***	1.6658***
	(1.2488)	(1.1979)	(0.0602)	(0.0602)	(0.0598)	(0.0598)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Chi-square	90.01	99.94	629.59	640.38	650.27	657.14
Log likelihood	−284.67	−278.91				
Number of observations	4667	4667	4667	4667	4667	4667
Overall R-squared			0.2936	0.2979	0.3016	0.3030
R-squared within			0.0250	0.0248	0.0249	0.0261
R-squared between			0.3907	0.3963	0.3992	0.3995

Notes: When excluding employee-related CSR and non-employee-related CSR from Model 5, the effects of employee creativity on organizational creativity remain significant ($\beta = 0.05$, $p = 0.06$).
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

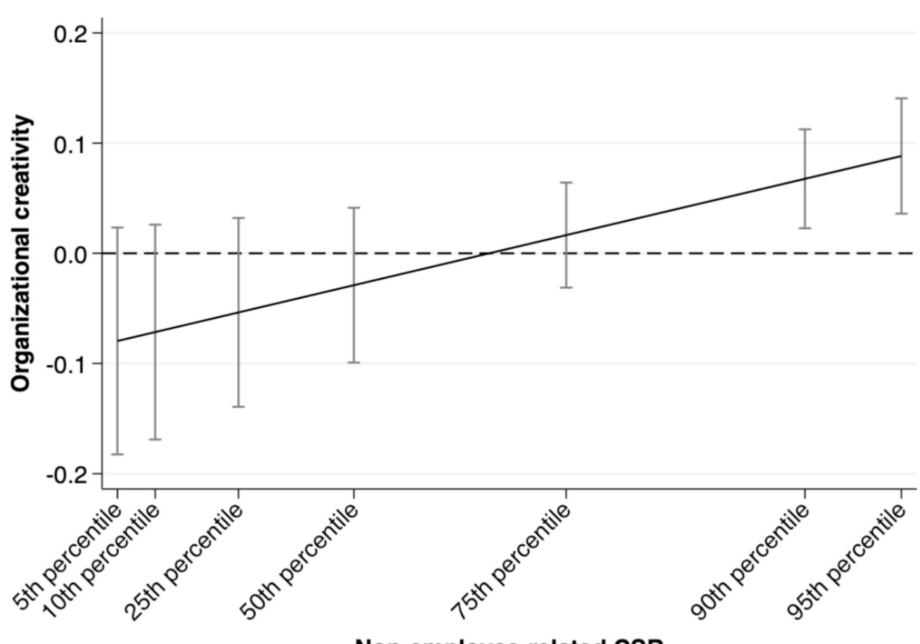

Fig. 2. Marginal effects of employee creativity on organizational creativity.

indirect effect of employee-related CSR on organizational creativity was 0.003 (p < 0.01; 95 % CI: 0.001–0.005). This finding suggests that employee creativity partially mediates the relationship between employee-related CSR and organizational creativity. Taken together, these tests and a cautious interpretation of the small decrease of the effect of employee-related CSR in Model 5 point to a partial mediation effect.

Hypothesis 3 states that non-employee-related CSR positively moderates the relationship between employee creativity and organizational creativity. To test this hypothesis, we introduce the interaction term between non-employee-related CSR and employee creativity in Model 6. The coefficient for the interaction term is statistically significant and positive ($\beta = 0.052$, $p < 0.05$), indicating that the relationship between employee creativity and organizational creativity is contingent on non-employee-related CSR. This evidence seems to suggest that Hypothesis 3 is supported. To better examine this relationship, we plotted the marginal effects of employee creativity on organizational creativity across various levels of non-employee-related CSR, as illustrated in Fig. 2. It reveals notable differences in the marginal effects of employee creativity at various levels of non-employee-related CSR, providing additional support for Hypothesis 3. The marginal effects of employee creativity on organizational creativity become statistically significant and positive at approximately the 80th percentile. This suggests that only a high level of non-employee-related CSR fosters the transformation from employee creativity to organizational creativity.

Hypothesis 4 predicted that non-employee-related CSR moderates the indirect and total effects of employee-related CSR on organizational creativity. To examine this hypothesis, we conducted moderated path analytic tests (Edwards and Lambert, 2007; Jia et al., 2014; Liao et al., 2010; Liu et al., 2011). The results in Table 3 show that the impact of employee-related CSR on organizational creativity, mediated by employee creativity, varies depending on the level of non-employee-related CSR. With high non-employee-related CSR, the indirect effect of employee-related CSR on organizational creativity is significant ($\beta = 0.003$, $p < 0.01$), as is the total effect of employee-related CSR on organizational creativity ($\beta = 0.050$, $p < 0.01$). These indirect effects are insignificant at low levels of non-employee-related CSR. Overall, these findings provide support for Hypothesis 4.

Additionally, we tested the significance of the indirect effect using the Karlson and Holm (2011) method. Our analysis reveals that 3 % of the total effect of employee-related CSR on organizational creativity is mediated by employee creativity for the average firm in the full sample. This relatively low percentage is expected for two reasons. First, our employee creativity measure focuses on top-performing employees who receive awards, thereby excluding those with more moderate creativity levels. The fact that this smaller group of employees already shows such a sizable mediation effect is particularly notable in this context. Second, the findings align with our conceptual model and moderation analysis, which demonstrates that the indirect effects are present only when non-employee-related CSR is high. Given this, we further investigated the significance of the indirect effect accounting for the conditional effect of non-employee-related CSR. Specifically, we further analyzed the direct versus indirect effects for firms with high non-employee-related CSR. For firms above the median in non-employee CSR (2334 observations), the indirect effects account for 29 % of the total effect. For those above the mean (1857 observations), the indirect effects increase to 42 %. Thus, the magnitude of indirect effects of high levels of non-employee-related CSR is in line with our expectations.

5.1. Robustness checks

We conducted a series of supplementary analyses to assess the robustness and reliability of our findings. First, to isolate firm-level effects and better control for industry-wide effects in our CSR measures, we introduced an alternative measure for both employee-related and non-employee-related CSR, adjusting our original measures using industry-average values based on the 2-digit TRBC code. The results, presented in Table A2 of the Online Appendix, support our main analysis. Second, we implemented an alternative estimation method in response to concerns about autocorrelation within observations of the same firm. We employed Generalized Estimation Equations (GEE) to test our hypotheses (Bliese, 2000; Certo and Semadeni, 2006). GEE models are adept at accounting for correlated error terms within firms (Liang and Zeger, 1986) and allow the specification of distribution family, link function, and correlation structure. The results are presented in Table A3 in the Online Appendix. In Models 1 and 2, we employed a binomial distribution, logit link function, and an exchangeable correlation structure, given that the dependent variable, employee creativity, is binary. For Models 3–6, we opted for a Gaussian distribution, identity link function, and maintained the exchangeable correlation structure. Again, the results are consistent with our main findings.

Moreover, even though the structure of our data suggests the use of a random effects model as previously noted, we ran hierarchical, i.e., mixed effects models which allow incorporating both fixed and random variance components (cf. Steinberg et al., 2023). By including fixed effects for time and random effects for industry and firm, the model captures advantages of both fixed-effects and random-effects approaches. This allows us to control for unobserved heterogeneity at the firm level while leveraging between-industry variation. The results (presented in Table A4 in the Online Appendix) supported our main findings.

Additionally, a potential endogeneity issue could bias our results. We proposed that employee creativity positively influences organizational creativity, but reverse causality is also possible—a creative organization might attract creative employees. To ensure that this potential endogeneity did not bias our results, we performed the two-step System Generalized Method of Moments (GMM) (Blundell and Bond, 1998). The System GMM combines differential GMM and horizontal GMM estimation methods to enhance parameter estimation efficiency. When employing instrumental variable (IV) estimation, it is necessary to assess the validity of the chosen instruments. Initially, Arellano and Bond's (1991) autocorrelation test was employed, revealing no second-order autocorrelation. Furthermore, we checked the validity of the instruments through the Hansen (1982) tests for over-identifying restrictions. The results, presented in Table 4, could not reject the null hypothesis, which provides support for the validity of all instrument variables selected in this model. Findings also show that the regression coefficients for employee creativity ($\beta = 0.063$, $p = 0.062$) and the interaction between employee creativity and non-employee-related CSR ($\beta = 0.051$, $p < 0.05$) are both significantly positive, indicating that reverse causality is not driving our results.

Lastly, we followed another approach to address potential reverse causality concerns. Regarding the relation between employee-related CSR and employee creativity, we follow Flammer and Luo's (2017) approach, using the difference-in-difference (DID) method to compare

Table 3
Moderated path analytic results—direct, indirect, and total effects of employee-related CSR on organizational creativity.

Moderator variable	Stage		Effect		
	First (IV to Mediator)	Second (Mediator to DV)	Direct	Indirect	Total
Low non-employee-related CSR	0.0081***	0.0694	0.0476***	0.0006	0.0481***
High non-employee-related CSR	0.0081***	0.3143***	0.0476**	0.0025***	0.0501**

*** p < 0.01, ** p < 0.05, * p < 0.10.

Table 4
The results of System GMM regressions.

	Organizational creativity	
	Model 1	Model 2
Organizational creativity$_{t-1}$	0.5192**	0.6810***
	(0.2105)	(0.1860)
Organizational creativity$_{t-2}$	0.0053	−0.1113
	(0.0956)	(0.1029)
Employee-related CSR	0.0220*	0.0228**
	(0.0114)	(0.0101)
Employee creativity	0.0626*	0.0318
	(0.0336)	(0.0469)
Employee creativity X Non-employee-related CSR		0.0513**
		(0.0221)
Control variables		
Non-employee-related CSR	0.0194*	0.0187*
	(0.0108)	(0.0104)
Firm size	−0.0358**	−0.0371***
	(0.0162)	(0.0139)
Tobin's Q	0.0599*	0.0532*
	(0.0348)	(0.0319)
Employee productivity	−0.0046	−0.0045
	(0.0042)	(0.0050)
Exploration orientation	0.0186**	0.0170**
	(0.0086)	(0.0085)
Differentiation strategy	0.0508***	0.0476***
	(0.0170)	(0.0148)
Slack resources	0.0570***	0.0527***
	(0.0184)	(0.0140)
Human resource slack	−0.0066*	−0.0060*
	(0.0038)	(0.0034)
Constant	0.8151***	0.7435***
	(0.2287)	(0.1767)
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
AR(1)	−1.90	−2.64
[P-value]	[0.06]	[0.01]
AR(2)	−1.06	−0.04
[P-value]	[0.29]	[0.97]
Hansen test	29.88	43.45
[P-value]	[0.23]	[0.70]
Number of observations	4022	4022

Notes: System GMM estimation, which incorporates lagged dependent variables, led to the loss of two years of data and resulted in 4022 observations.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5
Large increases in unemployment insurance benefits.

	Employee creativity			
	Model 1	Model 2	Model 3	Model 4
Large increases in UI benefits (Treatment x Period 0)	0.0015			
	(0.0130)			
Large increases in UI benefits (Treatment x Period 1)		−0.0079		
		(0.0208)		
Large increases in UI benefits (Treatment x Period 2)			0.0459**	
			(0.0226)	
Large increases in UI benefits (−1)				−0.0062
				(0.0138)
Large increases in UI benefits (0)				0.0010
				(0.0138)
Large increases in UI benefits (1)				−0.0106
				(0.0214)
Large increases in UI benefits (2)				0.0460**
				(0.0228)
Control variables				
Non-employee-related CSR	0.0005	0.0006	0.0004	0.0004
	(0.0057)	(0.0057)	(0.0057)	(0.0057)
Firm size	0.0226*	0.0224*	0.0220*	0.0219*
	(0.0132)	(0.0131)	(0.0131)	(0.0132)
Tobin's Q	−0.0138	−0.0140	−0.0137	−0.0142
	(0.0225)	(0.0225)	(0.0225)	(0.0226)
Employee productivity	0.0010	0.0010	0.0010	0.0010
	(0.0024)	(0.0024)	(0.0024)	(0.0024)
Constant	0.0101	0.0104	0.0075	0.0079
	(0.0100)	(0.0100)	(0.0101)	(0.0101)
Year fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
R-squared	0.0510	0.0509	0.0507	0.0507
Number of observations	4667	4667	4667	4667

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

creativity between firms in states with large increases in unemployment insurance (UI) benefits (treatment group) and those without such increases (control group). We build on Flammer and Luo's (2017) finding that significant UI benefit increases drive employee-related CSR, while they do not directly relate to our dependent variable. Building on this, we examine changes in employee creativity before and after significant increases in UI benefits.

The results (Table 5) show no significant effects in Models 1 and 2 for the year of or the year immediately following the treatment. However, in Model 3, the treatment effect is positive and significant in the second year post-increase. Model 4, which uses separate dummies for the year before (−1), the year of (0), and subsequent years (1 and 2), confirms no pre-existing trend and no effect in the year of and the first year after the treatment. A significant effect emerges in the second year after the treatment ($\beta = 0.046$, $p < 0.05$). Thus, these findings were consistent with our main analyses' results.

To address endogeneity concerns between employee and organizational creativity, we apply the same method used in the first stage of the model—the DID method—to compare the creativity of firms in states with significant increases in the average personal income tax rate (ATR) (treatment group) to those without such increases (control group). Moretti and Wilson (2017) showed that state tax changes impact the location of star scientists and skilled workers. This might mean that significant ATR increases drive skilled workers to relocate, reducing creative talent in that state and, consequently, creativity of firms headquartered there. ATR remains generally stable, with an average change of 0.0008 and a standard deviation of 0.01 across 51 states from 2007 to 2019. However, there is a noticeable discontinuity around the last decile (the largest 10 %), which corresponds to an increase of 0.02. Using this cutoff point, we define a "significant" ATR increase as 0.02 or more, which we expect to result in individual relocation.

Results (Table 6) show no significant effect in Models 1 and 2 for the year of or the year following the tax increase. However, the second year (Model 3) shows a negative and significant effect in line with our expectation that significant ATR changes would affect individual relocation with a moderate time lag. Model 4, analyzing dynamic effects, confirms no pre-existing trend and no significant effects in the year of or immediately after the increase. A significant negative effect emerges in the second year ($\beta = -0.046$, $p < 0.05$). Again, these findings increase our confidence in our results.

5.2. Supplementary analysis

Our moderation analysis shows that firms with high levels of non-employee-related CSR are better able to translate employee creativity into organizational creativity. However, the analysis also indicates that non-employee-related CSR does not have a significant direct effect on organizational creativity, while employee-related CSR is positively associated with it (see Table 2, Model 4). To further explore how highly creative organizations may differ in their CSR initiatives, we compared them to firms with lower creativity. We conducted a mean-comparison test (*t*-test) by splitting the sample at the median of organizational creativity, with the exploratory findings displayed in the Online Appendix (see Table A5). Interestingly, results show that firms with high organizational creativity have lower scores in the environmental and governance pillars compared to those with lower creativity, but they

Table 6
Large increases in state tax.

	Organizational creativity			
	Model 1	Model 2	Model 3	Model 4
Large increases in State tax (Treatment x Period 0)	−0.0095 (0.0197)			
Large increases in State tax (Treatment x Period 1)		−0.0057 (0.0196)		
Large increases in State tax (Treatment x Period 2)			−0.0456** (0.0183)	
Large increases in State tax (−1)				−0.0060 (0.0314)
Large increases in State tax (0)				−0.0095 (0.0197)
Large increases in State tax (1)				−0.0057 (0.0196)
Large increases in State tax (2)				−0.0456** (0.0183)
Control variables				
Employee-related CSR	0.0030 (0.0080)	0.0030 (0.0080)	0.0030 (0.0080)	0.0029 (0.0080)
Non-employee-related CSR	−0.0007 (0.0099)	−0.0007 (0.0099)	−0.0007 (0.0099)	−0.0007 (0.0099)
Firm size	−0.0474** (0.0211)	−0.0474** (0.0211)	−0.0474** (0.0211)	−0.0475** (0.0211)
Tobin's Q	−0.0530 (0.0356)	−0.0530 (0.0356)	−0.0530 (0.0356)	−0.0530 (0.0356)
Employee productivity	−0.0034 (0.0037)	−0.0034 (0.0037)	−0.0034 (0.0037)	−0.0034 (0.0037)
Exploration orientation	−0.0049 (0.0059)	−0.0049 (0.0059)	−0.0049 (0.0059)	−0.0049 (0.0059)
Differentiation strategy	0.0134 (0.0095)	0.0134 (0.0095)	0.0134 (0.0095)	0.0134 (0.0095)
Slack resources	0.0403*** (0.0082)	0.0403*** (0.0082)	0.0403*** (0.0082)	0.0403*** (0.0082)
Human resource slack	0.0004 (0.0045)	0.0004 (0.0045)	0.0004 (0.0045)	0.0004 (0.0045)
Constant	1.9105*** (0.0159)	1.9105*** (0.0159)	1.9105*** (0.0159)	1.9104*** (0.0159)
Year fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
R-squared	0.1511	0.1507	0.1507	0.1504
Number of observations	4667	4667	4667	4667

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

exhibit higher scores in the social pillar. This difference persists even after excluding employee-related items from the social pillar, focusing solely on non-employee-related components. Relating this to our theoretical framework, we posit that employees must experience meaningfulness in their work to fully commit to the organization's success. Tasks associated with social responsibility, such as volunteer projects, likely provide employees with a sense of meaningfulness 'in work' (Aguinis and Glavas, 2013; Carnahan et al., 2017; Pratt and Ashforth, 2003). The preliminary findings from our supplementary analysis hint at the possibility that this sense of meaningfulness is most effectively fostered by social CSR, as opposed to environmental or governance-related initiatives. As such, social CSR appears to be an especially relevant internal driver and context to facilitate the transformation of employee into organizational creativity.

6. Discussion

Our study aimed to explore the transformation of individual into organizational creativity and the role of diverse CSR initiatives. We proposed that employee-related CSR champions organizational creativity by nurturing employee creativity, and that non-employee-related CSR catalyzes the transformation of employee into organizational creativity. Our findings, which provide support for these theoretical expectations, are particularly important because creativity often serves as the cradle of innovation and is thus a pivotal force in the continuous evolution of organizations (Kauppila et al., 2018; Shao et al., 2019). Our research has several theoretical and practical implications.

6.1. Theoretical implications

Our study contributes to the literatures on the transformation of individual to organizational creativity and on the link between CSR and creativity in multiple ways. First, this study extends the literature on the intricate relationship between individual and organizational creativity by identifying CSR as a context that facilitates the transformation of the former into the latter. Prior research has identified a number of factors that facilitate this transformation, such as outward personnel mobility networks (Godart et al., 2014), high-performing work practices (Jeong and Shin, 2019), or individuals' leadership and entrepreneurial orientation (Gao et al., 2021); but this research has not fully accounted for the influence of organizational context on creativity transformation. Exploring this issue is relevant because a variety of contextual factors have been shown to impact each level of creativity independently (see, for example, Amabile and Pratt, 2016). We claimed, and found supporting evidence, that CSR, with its large array of activities directed towards employees but also towards other stakeholders, provides a context that supports the transformation of individual into organizational creativity. Based on the dichotomy of CSR activities–i.e., employee-related and non-employee-related CSR–our model theorized two mechanisms through which organizational context may influence the transformation of individual into organizational creativity, depending on the stakeholders towards which CSR activities are directed. While those aspects of organizational context that are directly related to employee satisfaction, engagement and motivation (i.e., employee-related CSR) impact their individual creativity, which in turn increases organizational creativity, aspects of organizational context that instill in employees a sense of belonging, meaningfulness and identification with the firm (i.e., non-employee-related CSR) influences organizational creativity indirectly by tapping into the potential of employees to work together and generate organizational outcomes. Our study is the first to theorize and empirically test distinct mechanisms through which organizational context influences creativity transformation.

Our theory and findings also add to existing CSR-creativity research in multiple ways. In contrast to previous studies on CSR and employee creativity that have treated CSR as a single construct and have not distinguished between different types of CSR (Abdelmotaleb et al., 2018; Brammer et al., 2015), we clarified that only employee-related CSR fosters employee creativity, while non-employee-related CSR has no statistically significant effect on employee creativity. This solidifies and extends Glavas and Piderit's (2009) claim that corporate citizenship increases individual creative engagement. By highlighting the effects of employee-related CSR on employee creativity, we substantiate the positive effect of CSR on individual performance.

Our study also substantiates and specifies the relationship between CSR and creativity. By establishing a connection between CSR and organizational creativity, our research provides a more complete picture of how CSR influences creativity in the workplace (Chaudhary and Akhouri, 2019; Glavas and Piderit, 2009; Hur et al., 2018). Previous research on the relationship between CSR and creativity has solely focused on employee creativity (Brammer et al., 2015; Chaudhary and Akhouri, 2019; Glavas and Piderit, 2009; Hur et al., 2018) and has given little attention to the link between CSR and organizational creativity. In this study, we found that CSR not only improves employee creativity but also organizational creativity, thereby complementing past studies on the link between CSR and creativity. In doing so, we delineate the specific types of CSR that are particularly advantageous in the transformation of individual into organizational creativity, elucidating the function of each. We reveal the importance of a formerly neglected type of CSR, non-employee-related CSR, in explaining that transformation. By engaging in this type of CSR, firms foster a sense of attachment to the organization among employees that will be reflected in organizational citizenship behaviors and increased cooperation with colleagues (Flammer and Luo, 2017; Pratt and Ashforth, 2003; Rupp et al., 2018), which will facilitate the sharing, building upon, and critiquing of novel ideas that will result in improved products, processes or forms of organization (Aguinis and Glavas, 2019; Chaudhary and Akhouri, 2018). Thus, our study offers to the literature a more holistic and nuanced perspective on the role of CSR in fostering organizational creativity.

Extending our previous arguments about the different types of CSR, in supplementary analyses we separated CSR into its environmental, social, and governance components. We found that more creative organizations score higher on the social pillar, suggesting another potentially important distinction among CSR types. Consistent with our findings, the higher score in the social pillar for creative organizations remains even after excluding employee-related items from the social pillar and only retaining the non-employee-related items. Construal level theory (Trope and Liberman, 2003, 2010) offers an explanation for this finding, as CSR activities that have a psychological impact on employees may be more likely to enhance meaningfulness at work and, in turn, may more effectively translate employee creativity into organizational creativity than activities under the environmental or governance pillars. For instance, governance-related CSR activities, such as executive compensation linked to CSR or the protection of shareholder rights, are less personally impactful to employees, limiting their perceived relevance. Although one might expect environmental initiatives to inspire purpose, they may still be perceived as more abstract compared to social initiatives, reducing their emotional resonance with employees. These nuances present promising avenues for future research to explore in more depth the mechanisms that drive these differences, particularly related to the role of meaningfulness at work (Aguinis and Glavas, 2013; Carnahan et al., 2017; Pratt and Ashforth, 2003).

Finally, we contribute to the literature on organizational creativity (Godart et al., 2014; Jeong and Shin, 2019; Lang and Lee, 2010) a novel and replicable measure of organizational creativity based on a textual analysis of corporate annual reports (10−*K*). Our text-based organizational creativity measure, which is distinct from previous innovation measures (Flammer and Kacperczyk, 2016; Heeley et al., 2007), provides a useful way to capture organizational creativity beyond tangible outcomes such as patents, namely creative ideas, procedures, and processes. A critical advantage of this measure is that it helps avoid confounding the concepts of creativity and innovation. Future studies on organizational creativity may build on our text-based organizational creativity measure.

6.2. Practical implications

To maintain competitiveness in a rapidly changing environment, organizations need to tap into their creativity. The findings of this study provide valuable insights into how to use CSR to boost both employee and organizational creativity. Our research indicates that different types of CSR have unique effects on creativity: CSR that directly supports employees stimulates their creativity; CSR unrelated to employees facilitates the transformation from employee into organizational creativity. Additional tests indicate that elements of the social pillar in CSR are particularly effective in driving these two effects. Overall, our findings suggest that CSR might be a valuable tool to promote both employee creativity as well as its transformation into organizational creativity. Therefore, firms interested in maximizing their creative potential should place greater emphasis on the adoption of employee-related CSR and formulate initiatives sensitive to employee needs and expectations. Further, in order to transform employee into organizational creativity, those firms should consider the interests of other stakeholders and engage in CSR that benefits them. Given the unique effects of various CSR types, companies should engage in CSR aligned with their specific goals, particularly when faced with resource constraints that make the simultaneous engagement in all types of CSR impractical.

6.3. Limitations and future research

Our study has shortcomings that offer opportunities for future research. A first avenue may be to address the limitations in our data. Our measurement of employee creativity relies on awards that employees have received, using a dichotomous variable. The awards are highly prestigious and rarely granted more than once a year, making instances of multiple awards within a single year uncommon. To account for this, we opted for a binary measure of employee creativity. Since the theoretical construct of individual creativity is typically conceptualized as continuous, ranging from low to high levels, but in our analysis, it is measured as a binary variable, this may not capture the full range of variation in the construct. Moreover, by relying on award data, our method primarily captures instances of exceptional employee creativity, potentially overlooking those with moderate or lower levels of creativity who still contribute to organizational creativity. To complement our study and despite the advantages of our measure, scholars should consider alternative and perhaps multiple measures of employee creativity to capture the effects of CSR more comprehensively at different creativity levels. For example, this could involve analyzing media references to employees' creative contributions, their social media activity, or internal employee evaluation documents.

Second, our data focuses on the high-tech industry, where creativity plays a crucial role. This suggests that the effects of CSR on creativity may be particularly pronounced in our context, and, therefore, our findings may be less applicable to firms in other industries. Future research should explore the boundary conditions of CSR effects on employee and organizational creativity and identify whether and to what extent these effects also manifest in other industries. It is necessary and promising to delve deeper into those conditions that help transform employee into organizational creativity. While our study delivers evidence of moderation and partial mediation effects of CSR on individual and organizational creativity, future research should further explore the contextual factors that promote or impede the transformation of employee into organizational creativity.

7. Conclusion

We distinguish between the effects of employee-related and non-employee-related CSR on organizational creativity. Our findings show that only employee-related CSR can directly improve employee creativity–a precursor of organizational creativity. The second component of CSR, non-employee-related CSR, does not directly affect employee creativity, yet it motivates creative employees to offer their innovative ideas and talents to their organizations, thereby moderating the relationship between creativity at the individual and collective level. Both employee-related and non-employee-related CSR favorably influence organizational creativity, but as we theorized and showed empirically, their effects are unique.

CRediT authorship contribution statement

He Li: Writing – original draft, Methodology, Formal analysis, Conceptualization, Data curation, Investigation, Project administration, Validation, Writing – review & editing. **Philip J. Steinberg:** Writing – review & editing, Supervision, Methodology, Conceptualization, Validation, Writing – original draft. **Hille C. Bruns:** Writing – review & editing, Supervision, Conceptualization, Writing – original draft. **Jordi A. Surroca:** Writing – review & editing, Supervision, Methodology,

Conceptualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix 1. Construction of the organizational creativity dictionary

Step 1: dictionary generated from the definition of creativity and scale items	Step 2: additional dictionary word stems developed using the inductive approach	Step 3: final organizational creativity dictionary
adaptable; applicable; autonomy; creativity; creative; explore; first; flexible; generate; ground-breaking; improvement; innovative; novel; opportunity; new idea; new way; new method; new invention; new product; new service; new work process; new operating procedure; new knowledge; new approach; original; risk; seek; try; useful; valuable	change; develop*; discover*; enhanc*; intellectual property; introduc*; launch*; new technology (technologies); next generation; patent (patents); promot*; proprietary; pursu*; releas*; reorganiz*; research; research and development (R&D); restructuring; transform*	creat*; develop*; discover*; enhanc*; improve*; innovat*; intellectual property; introduc*; launch*; new product(s); new technology (technologies); next generation; opportunity (opportunities); patent (patents); promot*; proprietary; pursu*; releas*; reorganiz*; research and development (R&D); research*; restructur*; seek (seeks, sought, seeking); transform*

Appendix 2. Examples of keyword occurrences in content analysis

Word stem	Example
Research and Development	We anticipate that our general and administrative expenses will increase in the future to support continued *research and development* activities and potential commercialization of any product candidates we identify and develop.
Discover*	Research and development expenses consist primarily of costs incurred for our research activities, including our drug *discovery* efforts and preclinical studies under our research programs...
Improve*	Revenue increased 7 % in 2013 from 2012 due to *improvement* in our retail solutions, hospitality, and emerging industries operating segments offset by declines in our financial services operating segment.
Introduc*	Our revenue growth is dependent on our ability to continually develop and *introduce* new products to meet the changing technology and performance requirements of our customers…
Research*	We expect our expenses to increase in connection with our ongoing activities, particularly as we further advance our current *research* programs and our preclinical development activities;

Appendix 3. Summary of validity of organizational and employee creativity

Variables	Organizational creativity	Employee creativity	Observations	Database
R&D intensity	0.47***		3980	Compustat
Patent/ total asset	0.13***		2840	US Patents by WRDS
Patent/number of employees	0.16**		2839	US Patents by WRDS
America's Most Innovative Companies ranking	0.56**		56	Fortune
Granted patent application	0.05***		2955	US Patents by WRDS
Economic importance				
Consumer price index	0.10***		2661	Kogan et al. (2017)
Nominal innovation value	0.09***		2661	Kogan et al. (2017)
Technological novelty				
Impact	0.13***		1367	Arts et al. (2021)
Novelty	0.31***		1367	Arts et al. (2021)
Inventor number		0.18***	3538	PatentsView database
Star inventor number (patent count)		0.15***	3538	PatentsView database
Star inventor number (citation count)		0.13***	2063	US Patents by WRDS

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix 4. Examples of individual creativity awards

Award name	Awarding agency	Award description
Aircraft Design Award	American Institute of Aeronautics and Astronautics	The Aircraft Design Award is presented to an individual or team for an original concept or career contributions leading to a significant advancement in aircraft design or design technology.
Industrial Research & Development Award	American Institute of Chemical Engineers	This award recognizes individuals or teams working in the industries served by chemical engineers, for innovation that has resulted in the successful commercial development of new products and/or new processes for making useful products.
ACS Award for Creative Invention	American Chemical Society	To recognize a single inventor for the successful application of research in chemistry and/or chemical engineering that contributes to the material prosperity and happiness of people.

(*continued on next page*)

(*continued*)

Award name	Awarding agency	Award description
ASM Award for Applied and Biotechnological Research	American Society for Microbiology	Recognizes an outstanding scientist with distinguished research achievements in the development of products, processes and technologies that have advanced the microbial sciences.
ELITE Tech-Know Geek	PM360	Innovators who spearhead technological breakthroughs that reshape the industry.
Environmental Excellence in Transportation (E2T) Award	Society of Automotive Engineers	This award recognizes an individual or groups of individuals who through their ingenuity and dedication make significant innovations in reducing the environmental impact caused by the transportation industry. These innovative achievements may occur in motorized vehicles for land, sea, air, and space in the areas of fuels, alternative propulsion methods, fuel usage, materials, energy usage, manufacturing methods, logistics support, as well as in education, training and improving public awareness.

Note: The full list of awards can be provided upon request.

Appendix 5. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.respol.2025.105286.

Data availability

The authors do not have permission to share data.

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